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THREAT HUNT REPORT

AZURE HONEYPOT INVESTIGATION

Threat Hunting and Network Forensic Analysis
of an Internet-Facing Linux Environment



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PREPARED FOR:

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01 EXECUTIVE SUMMARY

A threat hunting investigation was conducted against an internet-facing Azure honeynet environment monitored through Microsoft Sentinel and Microsoft Defender for Cloud.

The investigation identified over 7,000 unauthorized SSH authentication attempts originating from multiple global IP addresses targeting commonly used Linux administrative accounts including:

- root
- admin
- ubuntu
- debian
- pi

Several source IPs were validated through VirusTotal and were associated with known SSH brute-force activity and automated scanning behavior.

Analysis indicates the activity is consistent with:

- automated credential spraying,
- internet-wide SSH scanning,
- opportunistic brute-force campaigns.

No successful compromise was observed during the investigation period.

02 ENVIRONMENT SCOPE

Environment Scope

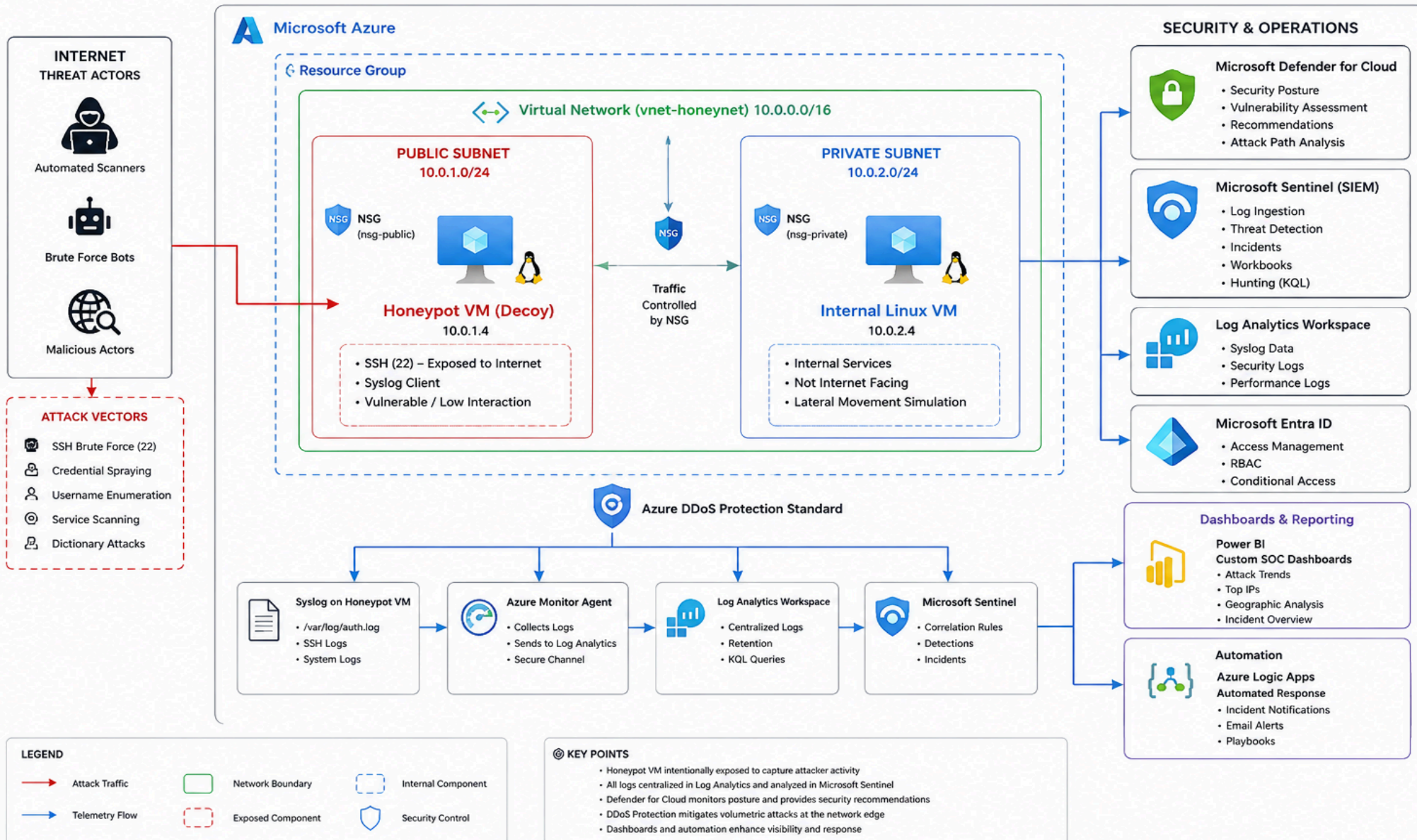
The monitored environment consisted of:

The primary objective of the investigation was to determine whether the email represented a legitimate communication or a malicious phishing attempt.



Azure Honeynet SOC Lab – Project Architecture

End-to-End Security Monitoring, Threat Detection & Response





Component	Purpose
Azure Virtual Network	Segmented lab network
Public-facing Linux VM	SSH honeynet decoy
Internal Linux VM	Simulated internal asset
NSG's	Traffic filtering
Azure DDoS Protection	Volumetric attack mitigation
Log Analytics Workspace	Telemetry ingestion
Microsoft Sentinel	SIEM and incident monitoring
Microsoft Defender for Cloud	Security posture management

The decoy VM intentionally exposed SSH services to the public internet to generate adversarial telemetry for monitoring and threat hunting purposes.

03 METHODOLOGY

Threat hunting activities were performed using Kusto Query Language (KQL) queries against Syslog telemetry collected within Microsoft Sentinel.

Analysis included:

- SSH authentication event monitoring
- source IP frequency analysis
- username targeting analysis
- geographic enrichment using GeoIP functions
- timeline-based attack trend analysis
- external threat intelligence validation using VirusTotal



EventTime [UTC] ↑↓	Facility	HostName	SeverityLevel	SyslogMessage	Type
> 5/27/2026, 10:34:38.000 AM	daemon	vm-internal-lab	info	motd-news.service: Deactivated...	Syslog
> 5/27/2026, 10:34:38.000 AM	daemon	vm-internal-lab	info	Finished motd-news.service - ...	Syslog
> 5/27/2026, 10:34:36.000 AM	daemon	vm-internal-lab	info	Starting motd-news.service - M...	Syslog
> 5/27/2026, 10:34:35.000 AM	auth	vm-decoy-lab	info	Received disconnect from 1.23...	Syslog
> 5/27/2026, 10:34:35.000 AM	auth	vm-decoy-lab	info	Disconnected from authenticati...	Syslog
> 5/27/2026, 10:34:31.000 AM	daemon	vm-internal-lab	info	2026-05-27T10:34:31.685#011L...	Syslog
> 5/27/2026, 10:34:26.000 AM	daemon	vm-internal-lab	info	2026-05-27T10:34:26.175#011L...	Syslog
> 5/27/2026, 10:34:26.000 AM	daemon	vm-internal-lab	info	2026-05-27T10:34:26.327#011L...	Syslog
> 5/27/2026, 10:34:25.000 AM	daemon	vm-decoy-lab	info	2026-05-27T10:34:25.063#011L...	Syslog

Data sources analyzed included:

- Syslog
- SecurityIncident
- Defender for Cloud recommendations

Network Forensic Analysis

Network traffic was captured using tcpdump on the internet-facing Linux honeynet VM. Captured packet data was exported in PCAP format and analyzed using Wireshark.

- SSH connection attempts,
- HTTP request inspection,
- reconnaissance activity,
- suspicious payload delivery attempts,
- network scanning behavior.

No.	Time	Source	Destination	Protocol	Length	Source Port	Destination Port	Info
22220	2026-05-27 11:26:23,828037Z	168.63.129.16	10.0.1.4	TCP	80	80	45398	[TCP Retransmission] 80 → 45398 [SYN, ACK] Seq=0 Ack=1 Win=65535 L...
22221	2026-05-27 11:26:23,828059Z	168.63.129.16	10.0.1.4	TCP	72	45398	80	45398 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=4082190338 TSec...
22222	2026-05-27 11:26:23,828060Z	168.63.129.16	10.0.1.4	TCP	72	45398	80	[TCP Dup ACK 22221#1] 45398 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0
22223	2026-05-27 11:26:23,828092Z	168.63.129.16	10.0.1.4	HTTP	272	45398	80	GET /machine/?comp=goalstate HTTP/1.1
22224	2026-05-27 11:26:23,828093Z	168.63.129.16	10.0.1.4	TCP	272	45398	80	[TCP Retransmission] 45398 → 80 [PSH, ACK] Seq=1 Ack=1 Win=64512 L...
22225	2026-05-27 11:26:23,828999Z	168.63.129.16	10.0.1.4	HTTP/X...	2369	80	45398	HTTP/1.1 200 OK
22226	2026-05-27 11:26:23,828999Z	168.63.129.16	10.0.1.4	TCP	2369	80	45398	[TCP Retransmission] 80 → 45398 [FIN, PSH, ACK] Seq=1 Ack=201 Win=...
22227	2026-05-27 11:26:23,829012Z	168.63.129.16	10.0.1.4	TCP	72	45398	80	45398 → 80 [ACK] Seq=201 Ack=2299 Win=69632 Len=0 TSval=4082190339
22228	2026-05-27 11:26:23,829013Z	168.63.129.16	10.0.1.4	TCP	72	45398	80	[TCP Dup ACK 22227#1] 45398 → 80 [ACK] Seq=201 Ack=2299 Win=69632
22229	2026-05-27 11:26:23,829201Z	168.63.129.16	10.0.1.4	TCP	72	45398	80	45398 → 80 [FIN, ACK] Seq=201 Ack=2299 Win=69632 Len=0 TSval=40821...
22230	2026-05-27 11:26:23,829202Z	168.63.129.16	10.0.1.4	TCP	72	45398	80	[TCP Retransmission] 45398 → 80 [FIN, ACK] Seq=201 Ack=2299 Win=69...
22231	2026-05-27 11:26:23,829243Z	168.63.129.16	10.0.1.4	TCP	72	80	45398	80 → 45398 [ACK] Seq=2299 Ack=202 Win=65280 Len=0 TSval=268877854
22232	2026-05-27 11:26:23,829243Z	168.63.129.16	10.0.1.4	TCP	72	80	45398	[TCP Dup ACK 22231#1] 80 → 45398 [ACK] Seq=2299 Ack=202 Win=65280
22233	2026-05-27 11:26:23,829715Z	168.63.129.16	10.0.1.4	TCP	80	33406	32526	33406 → 32526 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval...
22234	2026-05-27 11:26:23,829716Z	168.63.129.16	10.0.1.4	TCP	80	33406	32526	[TCP Retransmission] 33406 → 32526 [SYN] Seq=0 Win=64240 Len=0 MSS...
22235	2026-05-27 11:26:23,830144Z	168.63.129.16	10.0.1.4	TCP	80	32526	33406	32526 → 33406 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=2...
22236	2026-05-27 11:26:23,830144Z	168.63.129.16	10.0.1.4	TCP	80	32526	33406	[TCP Retransmission] 32526 → 33406 [SYN, ACK] Seq=0 Ack=1 Win=6553...
22237	2026-05-27 11:26:23,830157Z	168.63.129.16	10.0.1.4	TCP	72	33406	32526	33406 → 32526 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=4082190340 T...
22238	2026-05-27 11:26:23,830157Z	168.63.129.16	10.0.1.4	TCP	72	33406	32526	[TCP Dup ACK 22237#1] 33406 → 32526 [ACK] Seq=1 Ack=1 Win=64512 Le...
22239	2026-05-27 11:26:23,830180Z	168.63.129.16	10.0.1.4	HTTP	514	33406	32526	GET /vmSettings HTTP/1.1
22240	2026-05-27 11:26:23,830180Z	168.63.129.16	10.0.1.4	TCP	514	33406	32526	[TCP Retransmission] 33406 → 32526 [PSH, ACK] Seq=1 Ack=1 Win=6451...
22241	2026-05-27 11:26:23,833032Z	168.63.129.16	10.0.1.4	HTTP	248	32526	33406	HTTP/1.1 304 Not Modified

Observed files and payloads were extracted and validated using external threat intelligence sources including VirusTotal.



04 KEY FINDINGS

Total Attack Volume

A total of:

8100

SSH-related attack events were identified during the observation period.

Packet Analysis Findings

Packet inspection identified:

- repeated SSH brute-force attempts,
- automated scanning behavior,
- reconnaissance activity targeting exposed services,
- repeated authentication attempts against common Linux administrative usernames.

HTTP analysis identified:

- automated web probing activity,
- reconnaissance-based GET requests,
- scanner enumeration behavior.

Extracted files and transferred payloads were analyzed and hashed for threat intelligence validation. No confirmed malware payloads were identified during the observation period.



Most Active Source IPs

This indicates impersonation behavior commonly associated with phishing campaigns.

MOST AGGRESSIVE ATTACKERS	
SourceIP	Attempts
107.172.204.15	491
190.123.65.197	456
43.226.37.214	412
151.244.242.207	412
36.189.207.209	412
223.243.179.70	404
183.6.91.151	388
from 151.244.242.207	330
from 43.226.37.214	330
from 36.189.207.209	330
from 223.243.179.70	322

Credential Targeting Patterns

The most targeted usernames included:

- root
- admin
- ubuntu
- debian
- pi
- ftp

Additional targeted accounts included:

- ubnt
- AdminGPON
- orangepi
- zkagent
- Otrs
- User1
- nobody



SourceIP	Attempts	TargetedUsers
	8155	[""]
107.172.204.15	491	["admin","orange","root","test","user","cirros","rpc","user1","nobody","kali","linaro","postgres","openhbian","sshadmin",...]
190.123.65.197	456	["root","zkagent","dixi","backenderm","derrickml","admin","otrs","root0","root1","root2","root3","root4","rootalias"]
43.226.37.214	412	["root","user","ubuntu","debian","admin","pi","ftp"]
151.244.242.207	412	["root","user","ubuntu","debian","admin","pi","ftp"]
36.189.207.209	412	["root","user","ubuntu","debian","admin","pi","ftp"]
223.243.179.70	404	["root","user","ubuntu","debian","admin","pi","ftp"]
183.6.91.151	388	["root","user","ubuntu","debian","admin","pi","ftp"]
from 151.244.242.207	330	["user","ubuntu","debian","admin","pi","ftp"]
from 36.189.207.209	330	["user","ubuntu","debian","admin","pi","ftp"]

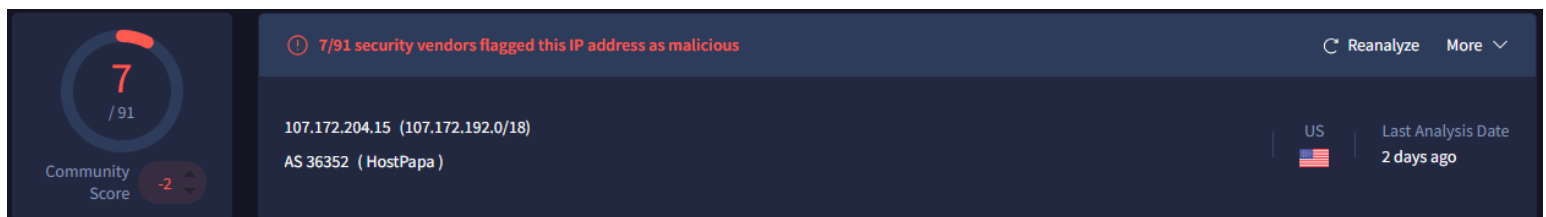
These patterns indicate attackers were attempting credential combinations associated with:

- Linux servers
- IoT infrastructure
- networking appliances
- embedded devices

05 THREAT ANALYSIS

Observed activity strongly aligned with:

- automated SSH brute-force campaigns,
- credential spraying,
- opportunistic internet-wide scanning.



Observed traffic patterns indicate predominantly opportunistic internet-wide reconnaissance and credential attack activity rather than targeted intrusion behavior.



Packet-level inspection showed no evidence of:

- successful malware delivery,
- exploit execution,
- lateral movement,
- persistence mechanisms,
- confirmed command-and-control communication.

Repeated username targeting patterns across multiple IP addresses suggest the use of:

- automated attack tooling,
- scripted credential dictionaries,
- botnet-based scanning infrastructure.

External validation through VirusTotal associated multiple source IPs with known malicious SSH brute-force behavior.

No evidence of successful authentication or lateral movement was identified during the analysis period. However, repeated credential targeting and scanning activity demonstrate continued exposure to automated attack infrastructure actively operating across the public internet.

06 MITRE ATT&CK MAPPING

MITRE ATT&CK Techniques Observed

Technique	ID	Description
Brute Force	T1110	Repeated credential attempts
Valid Accounts	T1078	Attempted use of common usernames
Network Service Scanning	T1046	SSH service probing
External Remote Services	T1133	Targeting internet-facing SSH

07 RISK ASSESSMENT

The environment was intentionally configured with internet-exposed SSH services for telemetry generation and threat analysis purposes.



Risks associated with this exposure included:

- credential brute-force attacks,
- service enumeration,
- unauthorized access attempts,
- opportunistic exploitation.

Compensating controls included:

- centralized logging,
- SIEM monitoring,
- incident detection,
- Defender for Cloud monitoring,
- network segmentation.

08 RECOMMENDATIONS

Recommended hardening measures include:

- restrict SSH exposure where operationally feasible,
- implement Just-In-Time VM access,
- enforce MFA for administrative access,
- enable endpoint detection and response (EDR),
- continuously monitor authentication events,
- maintain centralized SIEM telemetry collection,
- implement geo-based access restrictions where appropriate,
- regularly review Defender for Cloud recommendations.

09 CONCLUSION

The investigation successfully demonstrated the effectiveness of:

- Azure-native SIEM monitoring,
- cloud telemetry collection,
- threat hunting workflows,
- security posture management.

The honeynet environment generated realistic attack telemetry that enabled identification of automated SSH brute-force behavior and credential targeting activity across globally distributed infrastructure.